After reviewing this checklist, what will you do to become confident for all objectives?

3.3 Subtract Integers

Learning Objectives

By the end of this section, you will be able to:

- Model subtraction of integers
- Simplify expressions with integers
- Evaluate variable expressions with integers
- Translate words phrases to algebraic expressions
- Subtract integers in applications

BE PREPARED 3.4 Before you get started, take this readiness quiz.

Simplify: $12 - (8-1)$.
If you missed this problem, review Example 2.8.

BE PREPARED 3.5 Translate *the difference of 20 and -15* into an algebraic expression.
If you missed this problem, review Example 1.36.

BE PREPARED 3.6 Add: $-18 + 7$.
If you missed this problem, review Example 3.20.

Model Subtraction of Integers

Remember the story in the last section about the toddler and the cookies? Children learn how to subtract numbers through their everyday experiences. Real-life experiences serve as models for subtracting positive numbers, and in some cases, such as temperature, for adding negative as well as positive numbers. But it is difficult to relate subtracting negative numbers to common life experiences. Most people do not have an intuitive understanding of subtraction when negative numbers are involved. Math teachers use several different models to explain subtracting negative numbers.

We will continue to use counters to model subtraction. Remember, the blue counters represent positive numbers and the red counters represent negative numbers.

Perhaps when you were younger, you read $5 - 3$ as *five take away three*. When we use counters, we can think of subtraction the same way.

MANIPULATIVE MATHEMATICS

Doing the Manipulative Mathematics activity "Subtraction of Signed Numbers" will help you develop a better understanding of subtracting integers.

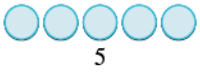
We will model four subtraction facts using the numbers 5 and 3.

$$5 - 3 \quad -5 - (-3) \quad -5 - 3 \quad 5 - (-3)$$

EXAMPLE 3.30

Model: $5 - 3$.

Solution

Interpret the expression.	$5 - 3$ means 5 take away 3.
Model the first number. Start with 5 positives.	 5

Take away the second number. So take away 3 positives.	
Find the counters that are left.	
	$5 - 3 = 2$. The difference between 5 and 3 is 2.

TRY IT 3.59 Model the expression:

$6 - 4$

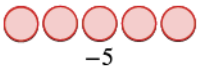
TRY IT 3.60 Model the expression:

$7 - 4$

EXAMPLE 3.31

Model: $-5 - (-3)$.

Solution

Interpret the expression.	$-5 - (-3)$ means -5 take away -3.
Model the first number. Start with 5 negatives.	
Take away the second number. So take away 3 negatives.	
Find the number of counters that are left.	
	$-5 - (-3) = -2$. The difference between -5 and -3 is -2.

TRY IT 3.61 Model the expression:

$-6 - (-4)$

TRY IT 3.62 Model the expression:

$-7 - (-4)$

Notice that Example 3.30 and Example 3.31 are very much alike.

- First, we subtracted 3 positives from 5 positives to get 2 positives.
- Then we subtracted 3 negatives from 5 negatives to get 2 negatives.

Each example used counters of only one color, and the "take away" model of subtraction was easy to apply.

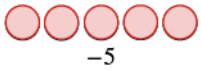
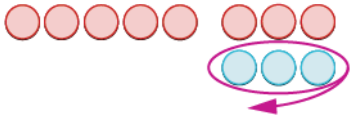


Now let's see what happens when we subtract one positive and one negative number. We will need to use both positive and negative counters and sometimes some neutral pairs, too. Adding a neutral pair does not change the value.

EXAMPLE 3.32

Model: $-5-3$.

Solution

Interpret the expression.	$-5-3$ means -5 take away 3.
Model the first number. Start with 5 negatives.	
Take away the second number. So we need to take away 3 positives.	
But there are no positives to take away. Add neutral pairs until you have 3 positives.	
Now take away 3 positives.	
Count the number of counters that are left.	
	$-5-3=-8$. The difference of -5 and 3 is -8.

TRY IT 3.63 Model the expression:

$-6-4$

TRY IT 3.64 Model the expression:

$-7-4$

EXAMPLE 3.33

Model: $5-(-3)$.

Solution

Interpret the expression.	$5 - (-3)$ means 5 take away -3.
Model the first number. Start with 5 positives.	
Take away the second number, so take away 3 negatives.	
But there are no negatives to take away. Add neutral pairs until you have 3 negatives.	
Then take away 3 negatives.	
Count the number of counters that are left.	 8 positives
	The difference of 5 and -3 is 8. $5 - (-3) = 8$

TRY IT 3.65 Model the expression:

$6 - (-4)$

TRY IT 3.66 Model the expression:

$7 - (-4)$

EXAMPLE 3.34

Model each subtraction.

(a) 8 − 2 (b) −5 − 4 (c) 6 − (−6) (d) −8 − (−3)

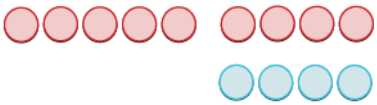
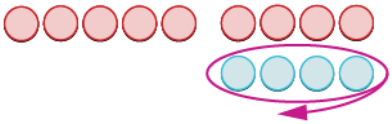
Solution

(a)

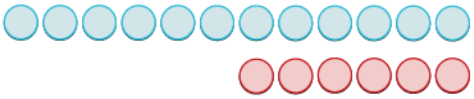
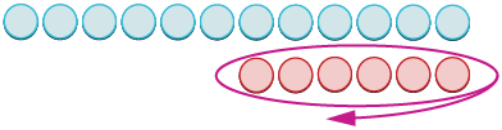
	$8 - 2$ This means 8 take away 2.
Start with 8 positives.	
Take away 2 positives.	

How many are left?	6
	$8 - 2 = 6$

ⓑ

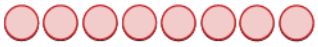
	$-5 - 4$ This means -5 take away 4.
Start with 5 negatives.	
You need to take away 4 positives. Add 4 neutral pairs to get 4 positives.	
Take away 4 positives.	
How many are left?	
	-9
	$-5 - 4 = -9$

ⓒ

	$6 - (-6)$ This means 6 take away -6.
Start with 6 positives.	
Add 6 neutrals to get 6 negatives to take away.	
Remove 6 negatives.	

How many are left?	
	12
	$6-(-6)=12$

ⓓ

	$-8-(-3)$ This means -8 take away -3.
Start with 8 negatives.	
Take away 3 negatives.	
How many are left?	
	-5
	$-8-(-3)=-5$

TRY IT 3.67 Model each subtraction.

ⓐ 7 - (-8) ⓑ -7 - (-2) ⓒ 4 - 1 ⓓ -6 - 8

TRY IT 3.68 Model each subtraction.

ⓐ 4 - (-6) ⓑ -8 - (-1) ⓒ 7 - 3 ⓓ -4 - 2

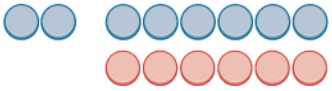
EXAMPLE 3.35

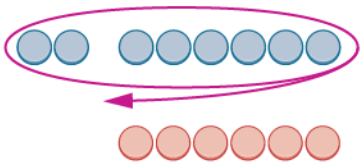
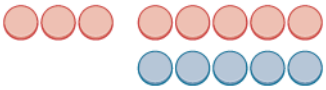
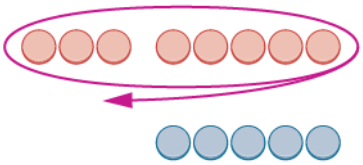
Model each subtraction expression:

ⓐ $2-8$ ⓑ $-3-(-8)$

Solution

ⓐ

We start with 2 positives.	 2
We need to take away 8 positives, but we have only 2.	
Add neutral pairs until there are 8 positives to take away.	

Then take away eight positives.	
Find the number of counters that are left. There are 6 negatives.	 6 negatives
	$2 - 8 = -6$
ⓑ We start with 3 negatives.	 3
We need to take away 8 negatives, but we have only 3.	
Add neutral pairs until there are 8 negatives to take away.	
Then take away the 8 negatives.	
Find the number of counters that are left. There are 5 positives.	 5 positives
	$-3 - (-8) = 5$

TRY IT 3.69 Model each subtraction expression.

ⓐ $7 - 9$ ⓑ $-5 - (-9)$

TRY IT 3.70 Model each subtraction expression.

ⓐ $4 - 7$ ⓑ $-7 - (-10)$

Simplify Expressions with Integers

Do you see a pattern? Are you ready to subtract integers without counters? Let's do two more subtractions. We'll think about how we would model these with counters, but we won't actually use the counters.

- Subtract $-23 - 7$.
  Think: We start with 23 negative counters.
  We have to subtract 7 positives, but there are no positives to take away.
  So we add 7 neutral pairs to get the 7 positives. Now we take away the 7 positives.
  So what's left? We have the original 23 negatives plus 7 more negatives from the neutral pair. The result is 30 negatives.

$$-23 - 7 = -30$$

Notice, that to subtract 7, we added 7 negatives.

- Subtract $30 - (-12)$.
  Think: We start with 30 positives.
  We have to subtract 12 negatives, but there are no negatives to take away.
  So we add 12 neutral pairs to the 30 positives. Now we take away the 12 negatives.
  What's left? We have the original 30 positives plus 12 more positives from the neutral pairs. The result is 42 positives.

$$30 - (-12) = 42$$

Notice that to subtract -12, we added 12.

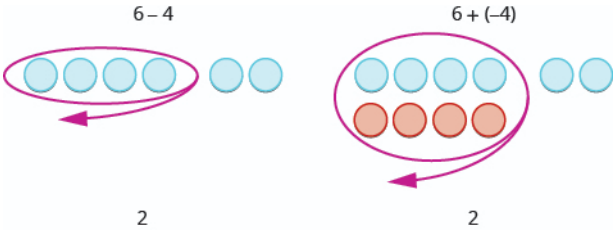
While we may not always use the counters, especially when we work with large numbers, practicing with them first gave us a concrete way to apply the concept, so that we can visualize and remember how to do the subtraction without the counters.

Have you noticed that subtraction of signed numbers can be done by adding the opposite? You will often see the idea, the Subtraction Property, written as follows:

Subtraction Property

$$a - b = a + (-b)$$

Look at these two examples.



We see that $6 - 4$ gives the same answer as $6 + (-4)$.

Of course, when we have a subtraction problem that has only positive numbers, like the first example, we just do the subtraction. We already knew how to subtract $6 - 4$ long ago. But knowing that $6 - 4$ gives the same answer as $6 + (-4)$ helps when we are subtracting negative numbers.

EXAMPLE 3.36

Simplify:

ⓐ $13 - 8$ and $13 + (-8)$ ⓑ $-17 - 9$ and $-17 + (-9)$

Solution

ⓐ

	$13 - 8$ and $13 + (-8)$
Subtract to simplify.	$13 - 8 = 5$
Add to simplify.	$13 + (-8) = 5$
Subtracting 8 from 13 is the same as adding −8 to 13.	

ⓑ

	$-17-9$ and $-17+(-9)$
Subtract to simplify.	$-17-9=-26$
Add to simplify.	$-17+(-9)=-26$
Subtracting 9 from −17 is the same as adding −9 to −17.	

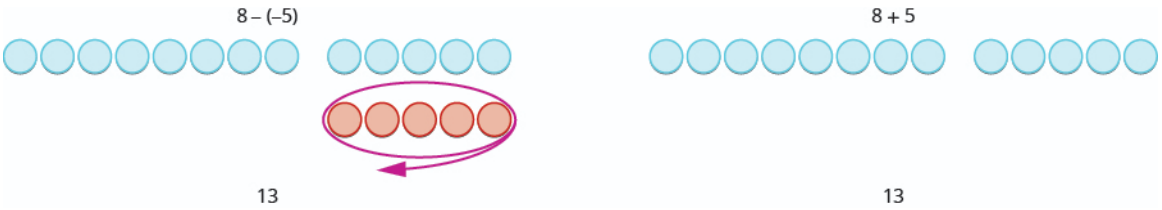
TRY IT 3.71 Simplify each expression:

ⓐ $21-13$ and $21+(-13)$ ⓑ $-11-7$ and $-11+(-7)$

TRY IT 3.72 Simplify each expression:

ⓐ $15-7$ and $15+(-7)$ ⓑ $-14-8$ and $-14+(-8)$

Now look what happens when we subtract a negative.



We see that $8-(-5)$ gives the same result as $8+5$. Subtracting a negative number is like adding a positive.

EXAMPLE 3.37

Simplify:

ⓐ $9-(-15)$ and $9+15$ ⓑ $-7-(-4)$ and $-7+4$

Solution

ⓐ

	$9-(-15)$ and $9+15$
Subtract to simplify.	$9-(-15)=24$
Add to simplify.	$9+15=24$
Subtracting −15 from 9 is the same as adding 15 to 9.	

ⓑ

	$-7-(-4)$ and $-7+4$
Subtract to simplify.	$-7-(-4)=-3$

Add to simplify.	$-7+4=-3$
Subtracting −4 from −7 is the same as adding 4 to −7	

TRY IT 3.73 Simplify each expression:

ⓐ $6-(-13)$ and $6+13$ ⓑ $-5-(-1)$ and $-5+1$

TRY IT 3.74 Simplify each expression:

ⓐ $4-(-19)$ and $4+19$ ⓐ $-4-(-7)$ and $-4+7$

Look again at the results of Example 3.30 - Example 3.33.

5− 3	−5− (−3)
2	−2
2 positives	2 negatives
When there would be enough counters of the color to take away, subtract.	
−5− 3	5− (−3)
−8	8
5 negatives, want to subtract 3 positives	5 positives, want to subtract 3 negatives
need neutral pairs	need neutral pairs
When there would not be enough of the counters to take away, add neutral pairs.	

Table 3.4 Subtraction of Integers

EXAMPLE 3.38

Simplify: $-74-(-58)$.

Solution

We are taking 58 negatives away from 74 negatives.	$-74-(-58)$
Subtract.	-16

TRY IT 3.75 Simplify the expression:

$-67-(-38)$

TRY IT 3.76 Simplify the expression:

$-83-(-57)$

EXAMPLE 3.39

Simplify: $7 - (-4 - 3) - 9$.

Solution

We use the order of operations to simplify this expression, performing operations inside the parentheses first. Then we subtract from left to right.

Simplify inside the parentheses first.	$7 - (-4 - 3) - 9$
Subtract from left to right.	$7 - (-7) - 9$
Subtract.	$14 - 9$
	5

TRY IT 3.77 Simplify the expression:

$8 - (-3 - 1) - 9$

TRY IT 3.78 Simplify the expression:

$12 - (-9 - 6) - 14$

EXAMPLE 3.40

Simplify: $3 \cdot 7 - 4 \cdot 7 - 5 \cdot 8$.

Solution

We use the order of operations to simplify this expression. First we multiply, and then subtract from left to right.

Multiply first.	$3 \cdot 7 - 4 \cdot 7 - 5 \cdot 8$
Subtract from left to right.	$21 - 28 - 40$
Subtract.	$-7 - 40$
	-47

TRY IT 3.79 Simplify the expression:

$6 \cdot 2 - 9 \cdot 1 - 8 \cdot 9.$

TRY IT 3.80 Simplify the expression:

$2 \cdot 5 - 3 \cdot 7 - 4 \cdot 9$

Evaluate Variable Expressions with Integers

Now we'll practice evaluating expressions that involve subtracting negative numbers as well as positive numbers.

EXAMPLE 3.41

Evaluate $x - 4$ when

ⓐ $x = 3$ ⓑ $x = -6$.

Solution

ⓐ To evaluate $x - 4$ when $x = 3$, substitute 3 for x in the expression.

	$x - 4$
Substitute 3 for x.	$3 - 4$
Subtract.	-1

ⓑ To evaluate $x - 4$ when $x = -6$, substitute -6 for x in the expression.

	$x - 4$
Substitute -6 for x.	$-6 - 4$
Subtract.	-10

TRY IT 3.81 Evaluate each expression:

$y - 7$ when

ⓐ $y = 5$ ⓑ $y = -8$

TRY IT 3.82 Evaluate each expression:

$m - 3$ when

ⓐ $m = 1$ ⓑ $m = -4$

EXAMPLE 3.42

Evaluate $20 - z$ when

ⓐ $z = 12$ ⓑ $z = -12$

Solution

ⓐ To evaluate $20 - z$ when $z = 12$, substitute 12 for z in the expression.

	$20 - z$
Substitute 12 for z.	$20 - 12$
Subtract.	8

ⓑ To evaluate $20 - z$ when $z = -12$, substitute -12 for z in the expression.

	$20 - z$
Substitute -12 for z.	$20 - (-12)$
Subtract.	32

TRY IT 3.83 Evaluate each expression:

$17 - k$ when

ⓐ $k = 19$ ⓑ $k = -19$

TRY IT 3.84 Evaluate each expression:

$-5 - b$ when

ⓐ $b = 14$ ⓑ $b = -14$

Translate Word Phrases to Algebraic Expressions

When we first introduced the operation symbols, we saw that the expression $a - b$ may be read in several ways as shown below.

$a - b$
a minus b
the difference of a and b
subtract b from a
b subtracted from a
b less than a

Figure 3.18

Be careful to get a and b in the right order!

EXAMPLE 3.43

Translate and then simplify:

ⓐ the difference of 13 and -21 ⓑ subtract 24 from -19

Solution

ⓐ A *difference* means subtraction. Subtract the numbers in the order they are given.

	the difference of 13 and -21
Translate.	$13 - (-21)$
Simplify.	34

ⓑ *Subtract* means to take 24 away from -19.

	subtract 24 from −19
Translate.	−19 − 24
Simplify.	−43

TRY IT 3.85 Translate and simplify:

ⓐ the difference of 14 and −23 ⓑ subtract 21 from −17

TRY IT 3.86 Translate and simplify:

ⓐ the difference of 11 and −19 ⓑ subtract 18 from −11

Subtract Integers in Applications

It's hard to find something if we don't know what we're looking for or what to call it. So when we solve an application problem, we first need to determine what we are asked to find. Then we can write a phrase that gives the information to find it. We'll translate the phrase into an expression and then simplify the expression to get the answer. Finally, we summarize the answer in a sentence to make sure it makes sense.

HOW TO

Solve Application Problems.

Step 1. Identify what you are asked to find.
Step 2. Write a phrase that gives the information to find it.
Step 3. Translate the phrase to an expression.
Step 4. Simplify the expression.
Step 5. Answer the question with a complete sentence.

EXAMPLE 3.44

In the morning, the temperature in Urbana, Illinois was 11 degrees Fahrenheit. By mid-afternoon, the temperature had dropped to −9 degrees Fahrenheit. What was the difference between the morning and afternoon temperatures?

Solution

Step 1. Identify what we are asked to find.	the difference between the morning and afternoon temperatures
Step 2. Write a phrase that gives the information to find it.	the difference of 11 and −9
Step 3. Translate the phrase to an expression. The word *difference* indicates subtraction.	11 − (−9)
Step 4. Simplify the expression.	20
Step 5. Write a complete sentence that answers the question.	The difference in temperature was 20 degrees Fahrenheit.

TRY IT 3.87 In the morning, the temperature in Anchorage, Alaska was 15 degrees Fahrenheit. By mid-afternoon the temperature had dropped to 30 degrees below zero. What was the difference between the morning and afternoon temperatures?

TRY IT 3.88 The temperature in Denver was −6 degrees Fahrenheit at lunchtime. By sunset the temperature had dropped to −15 degree Fahrenheit. What was the difference between the lunchtime and sunset temperatures?

Geography provides another application of negative numbers with the elevations of places below sea level.

EXAMPLE 3.45

Dinesh hiked from Mt. Whitney, the highest point in California, to Death Valley, the lowest point. The elevation of Mt. Whitney is 14,497 feet above sea level and the elevation of Death Valley is 282 feet below sea level. What is the difference in elevation between Mt. Whitney and Death Valley?

Solution

Step 1. What are we asked to find?	The difference in elevation between Mt. Whitney and Death Valley
Step 2. Write a phrase.	elevation of Mt. Whitney–elevation of Death Valley
Step 3. Translate.	$14{,}497 - (-282)$
Step 4. Simplify.	14,779
Step 5. Write a complete sentence that answers the question.	The difference in elevation is 14,779 feet.

TRY IT 3.89 One day, John hiked to the 10,023 foot summit of Haleakala volcano in Hawaii. The next day, while scuba diving, he dove to a cave 80 feet below sea level. What is the difference between the elevation of the summit of Haleakala and the depth of the cave?

TRY IT 3.90 The submarine Nautilus is at 340 feet below the surface of the water and the submarine Explorer is 573 feet below the surface of the water. What is the difference in the position of the Nautilus and the Explorer?

Managing your money can involve both positive and negative numbers. You might have overdraft protection on your checking account. This means the bank lets you write checks for more money than you have in your account (as long as they know they can get it back from you!)

EXAMPLE 3.46

Leslie has $25 in her checking account and she writes a check for $8.

(a) What is the balance after she writes the check?
(b) She writes a second check for $20. What is the new balance after this check?
(c) Leslie's friend told her that she had lost a check for $10 that Leslie had given her with her birthday card. What is the balance in Leslie's checking account now?

Solution

ⓐ

What are we asked to find?	The balance of the account
Write a phrase.	$25 minus $8
Translate	$25 − $8
Simplify.	$17
Write a sentence answer.	The balance is $17.

ⓑ

What are we asked to find?	The new balance
Write a phrase.	$17 minus $20
Translate	$17 − $20
Simplify.	−$3
Write a sentence answer.	She is overdrawn by $3.

ⓒ

What are we asked to find?	The new balance
Write a phrase.	$10 more than −$3
Translate	−$3 + $10
Simplify.	$7
Write a sentence answer.	The balance is now $7.

TRY IT 3.91 Araceli has $75 in her checking account and writes a check for $27.

ⓐ What is the balance after she writes the check?
ⓑ She writes a second check for $50. What is the new balance?
ⓒ The check for $20 that she sent a charity was never cashed. What is the balance in Araceli's checking account now?

TRY IT 3.92 Genevieve's bank account was overdrawn and the balance is −$78.

ⓐ She deposits a check for $24 that she earned babysitting. What is the new balance?
ⓑ She deposits another check for $49. Is she out of debt yet? What is her new balance?

LINKS TO LITERACY

The Links to Literacy activity "Elevator Magic" will provide you with another view of the topics covered in this section.

MEDIA

ACCESS ADDITIONAL ONLINE RESOURCES

Adding and Subtracting Integers (https://openstaxcollege.org/l/24AddSubtrInteg)
Subtracting Integers with Color Counters (https://openstaxcollege.org/l/24Subtrinteger)
Subtracting Integers Basics (https://openstaxcollege.org/l/24Subtractbasic)
Subtracting Integers (https://openstaxcollege.org/l/24introintegerr)
Integer Application (https://openstaxcollege.org/l/24integerappp)

SECTION 3.3 EXERCISES

Practice Makes Perfect

Model Subtraction of Integers

In the following exercises, model each expression and simplify.

127. $8-2$

128. $9-3$

129. $-5-(-1)$

130. $-6-(-4)$

131. $-5-4$

132. $-7-2$

133. $8-(-4)$

134. $7-(-3)$

Simplify Expressions with Integers

In the following exercises, simplify each expression.

135. (a) $15-6$ (b) $15+(-6)$

136. (a) $12-9$ (b) $12+(-9)$

137. (a) $44-28$ (b) $44+(-28)$

138. (a) $35-16$ (b) $35+(-16)$

139. (a) $8-(-9)$ (b) $8+9$

140. (a) $4-(-4)$ (b) $4+4$

141. (a) $27-(-18)$ (b) $27+18$

142. (a) $46-(-37)$ (b) $46+37$

In the following exercises, simplify each expression.

143. $15-(-12)$

144. $14-(-11)$

145. $10-(-19)$

146. $11-(-18)$

147. $48-87$

148. $45-69$

149. $31-79$

150. $39-81$

151. $-31-11$

152. $-32-18$

153. $-17-42$

154. $-19-46$

155. $-103-(-52)$

156. $-105-(-68)$

157. $-45-(-54)$

158. $-58-(-67)$

159. $8-3-7$

160. $9-6-5$

161. $-5-4+7$

162. $-3-8+4$

163. $-14-(-27)+9$

164. $-15-(-28)+5$

165. $71+(-10)-8$

166. $64+(-17)-9$

167. $-16-(-4+1)-7$

168. $-15-(-6+4)-3$

169. $(2-7)-(3-8)$

170. $(1-8)-(2-9)$

171. $-(6-8)-(2-4)$

172. $-(4-5)-(7-8)$

173. $25-[10-(3-12)]$

174. $32-[5-(15-20)]$

175. $6\cdot 3-4\cdot 3-7\cdot 2$

176. $5 \cdot 7 - 8 \cdot 2 - 4 \cdot 9$

177. $5^2 - 6^2$

178. $6^2 - 7^2$

Evaluate Variable Expressions with Integers

In the following exercises, evaluate each expression for the given values.

179. $x - 6$ when
ⓐ $x = 3$ ⓑ $x = -3$

180. $x - 4$ when
ⓐ $x = 5$ ⓑ $x = -5$

181. $5 - y$ when
ⓐ $y = 2$ ⓑ $y = -2$

182. $8 - y$ when
ⓐ $y = 3$ ⓑ $y = -3$

183. $4x^2 - 15x + 1$ when $x = 3$

184. $5x^2 - 14x + 7$ when $x = 2$

185. $-12 - 5x^2$ when $x = 6$

186. $-19 - 4x^2$ when $x = 5$

Translate Word Phrases to Algebraic Expressions

In the following exercises, translate each phrase into an algebraic expression and then simplify.

187. ⓐ The difference of 3 and -10
ⓑ Subtract -20 from 45

188. ⓐ The difference of 8 and -12
ⓑ Subtract -13 from 50

189. ⓐ The difference of -6 and 9
ⓑ Subtract -12 from -16

190. ⓐ The difference of -8 and 9
ⓑ Subtract -15 from -19

191. ⓐ 8 less than -17
ⓑ -24 minus 37

192. ⓐ 5 less than -14
ⓑ -13 minus 42

193. ⓐ 21 less than 6
ⓑ 31 subtracted from -19

194. ⓐ 34 less than 7
ⓑ 29 subtracted from -50

Subtract Integers in Applications

In the following exercises, solve the following applications.

195. **Temperature** One morning, the temperature in Urbana, Illinois, was $28°$ Fahrenheit. By evening, the temperature had dropped $38°$ Fahrenheit. What was the temperature that evening?

196. **Temperature** On Thursday, the temperature in Spincich Lake, Michigan, was $22°$ Fahrenheit. By Friday, the temperature had dropped $35°$ Fahrenheit. What was the temperature on Friday?

197. **Temperature** On January 15, the high temperature in Anaheim, California, was $84°$ Fahrenheit. That same day, the high temperature in Embarrass, Minnesota was $-12°$ Fahrenheit. What was the difference between the temperature in Anaheim and the temperature in Embarrass?

198. **Temperature** On January 21, the high temperature in Palm Springs, California, was $89°$, and the high temperature in Whitefield, New Hampshire was $-31°$. What was the difference between the temperature in Palm Springs and the temperature in Whitefield?

199. **Football** At the first down, the Warriors football team had the ball on their 30-yard line. On the next three downs, they gained 2 yards, lost 7 yards, and lost 4 yards. What was the yard line at the end of the third down?

200. **Football** At the first down, the Barons football team had the ball on their 20-yard line. On the next three downs, they lost 8 yards, gained 5 yards, and lost 6 yards. What was the yard line at the end of the third down?